

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech.)
SEM - I University Examination, May- 2013
M.Tech. (CAD/CAM)

ME – 703.01 Advanced Dynamics of Machines

Date: 10.05.2013, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION-I

- Q.1 (a) ABCD is a four bar mechanism. Link length AB = 50 mm, BC = 90 mm, CD = 110 mm, Fixed link DA = 125 mm and if the link AB rotates at constant angular velocity of 12 rad/sec. Determine velocity and acceleration of other links. 11
- Q.2 (a) Refer Figure.1. Determine by analytical method, the couple on crank 2 to be applied for equilibrium of the system, when a force of 500N acts on connecting rod at point C as shown in figure. Also determine the resultant of forces exerted on the frame of the engine. 12

OR

- Q.2 (a) Explain virtual work principle. Figure.2 shows an quick return mechanism. The driving crank OA is 60 mm long and rotates at a uniform speed of 200 r.p.m. in c.w. direction. For the position shown find torque required on crank for static equilibrium by using virtual work method if the force acting on slider R = 100N. 08
- (b) Explain the consideration of slider friction and turning friction in static force analysis. 04
- Q.3 (a) Find the inertia force for the following data of an I.C. engine. 02
Bore = 175mm, stroke = 200mm, engine speed = 500 r.p.m., length of connecting rod = 400 mm, crank angle = 60° from T.D.C. and mass of reciprocating parts = 180 kg.
- (b) A vertical petrol engine 100 mm diameter and 120 mm stroke has a connecting rod 250 mm long. The mass of the piston is 1.1 kg. the speed is 2000 r.p.m. On the expansion stroke with a crank 20° from T.D.C., the gas pressure is 700 kN/m². Determine 1. Net force on the piston, 2. Resultant load on the gudgeon pin, 3. Thrust on the cylinder walls. 4. speed above which, the other things remaining same, the gudgeon pin load will be reversed in direction (refer Figure.3). 04
- (c) i) Explain cam curvature and undercutting in roller followers. 03
ii) Explain ramp of the Cam or pre-Cam 03

OR

- Q.3 (a) Explain dynamical equivalent system. A connecting rod is suspended a point 25 mm above the centre of small end and 650 mm above its CG, its mass being 37.5kg. When permitted to oscillate, the time period is found to be 1.87 seconds. Find the dynamical equivalent system constituted of two masses, one of which is located at the small end centre. 04
- (b) An eccentric plate cam of 160 mm diameter and eccentricity of 40mm provides motion to a spring loaded follower of mass 2Kg whose axis is perpendicular to the axis of the cam and passes through its center. The spring has stiffness of 25N/mm. It is found that at certain speed, the follower ceases to have contact with cam, when cam has moved through 110° from its lowest position. Determine this speed. The initial compression of the spring is 32mm. Determine also the limiting angular speed of the cam to avoid cam jump 08

SECTION- II

- Q.4 (a)** Define degree of freedom of mechanism. Explain the mobility of spatial mechanism. **07**
- (b)** Three Position Synthesis – 4 bar mechanism by Inversion method. **04**
- Q.5 (a)** Explain i) Path Generation ii) Function generation methods of synthesis. **04**
- (b)** Explain Coupler curve synthesis for 4-bar mechanism **04**
- (c)** Synthesize a four-bar linkage in one of its positions, it satisfy the following values **04**
for the angular velocities and accelerations.
 $\omega_2 = 20 \text{ rad/sec}, \omega_3 = 08 \text{ rad/sec}, \omega_4 = 10 \text{ rad/sec}$
 $\text{and } \alpha_2 = 0 \text{ rad/sec}^2, \alpha_3 = 100 \text{ rad/sec}^2, \alpha_4 = -150 \text{ rad/sec}^2.$

OR

- Q.5 (a)** Define precision points in Chebyshev method of synthesis. Explain the graphical **12**
method to find the chebyshev precision points. Design and draw a slider crank
mechanism in which displacement of slider S is given by following relation :

$$S_j = \frac{66825 + 630\theta_j - 7\theta_j^2}{8100}$$

Where θ_j = Crank angle of rotation in the range, $45^\circ \leq \theta \leq 135^\circ$. Use three point Chebyshev spacing.

- Q.6 (a)** Explain balancing of In-line Engine. And investigate the state of primary and **08**
secondary balancing of a 4-stroke cycle four cylinder engine with firing order I-II-
III-IV. What will be the change in firing-order is altered to I-II-IV-III?
- (b)** Why multi-cylinder engines are used? Discuss. **04**

OR

- Q.6 (a)** Explain active and passive balancing. **02**
- (b)** Discuss on balancing of resultant primary and secondary forces for V-Engine. What **10**
is the value of resultant primary force for 60° V-Engine in terms mass, angular speed,
crank radius and angle θ .

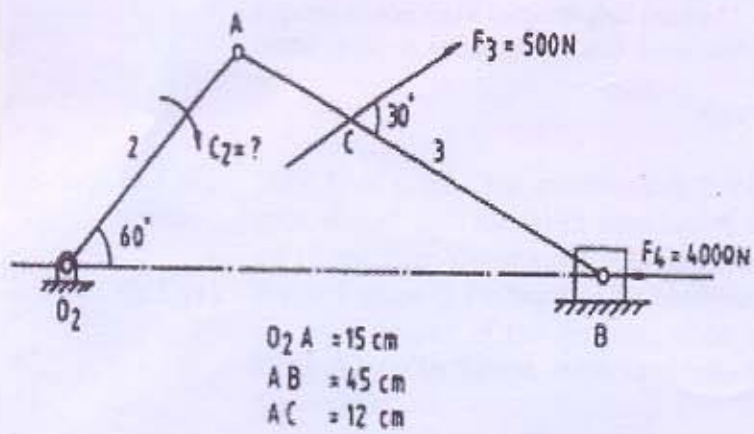


Figure. 1

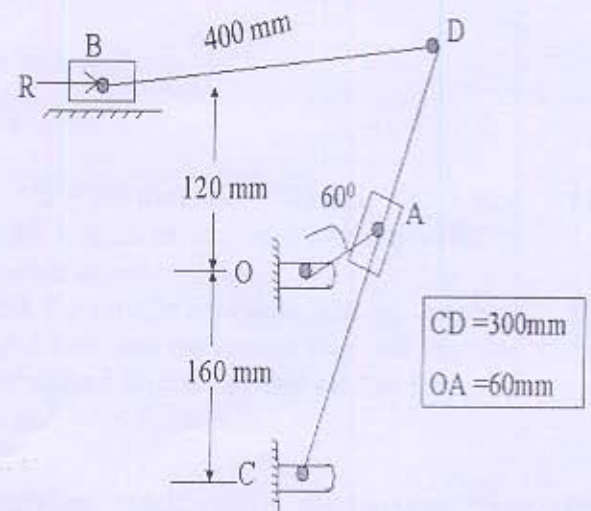


Figure. 2



Figure.3